



AI-POWERED EDUCATIONAL CHATBOT



INTRODUCTION



1. **AI-Powered Educational Chabot** designed to provide accurate answers of student's educational questions.
2. Uses a **CSV-based knowledge base** containing approximately 300 question-answer pairs.
3. Uses **Sentence Transformer and semantic similarity** to understand the meaning of user questions.
4. Provides an interactive **Stream lit Chabot with persistent SQLite chat history**.



OBJECTIVE

1. To develop an intelligent Chabot for **educational question answering**.
2. To retrieve relevant answers using **Sentence Transformer embedding and Cosine Similarity**.
3. To provide a **user-friendly Stream lit interface** with suggested questions and similarity scores.
4. To maintain **persistent chat history using SQLite** across refreshes and application restarts.





METHODOLOGY

1. **Dataset Collection:** Prepare around 300 educational question-answer pairs in a CSV knowledge base.
2. **Data Cleaning:** Remove extra spaces, duplicates, empty values, and unnecessary formatting.
3. **Data Selection:** Use only the question and answer columns for the Chabot.
4. **Sentence Transformer:** Use all-MiniLM-L6-v2 to understand the semantic meaning of questions.
5. **Embedding Generation:** Convert all knowledge-base questions into numerical vector embedding.
6. **User Query:** Convert the user's question into an embedding using the same Sentence Transformer.
7. **Cosine Similarity:** Compare the user-query embedding with all stored question embedding.
8. **Best Match:** Select the question having the highest similarity score.
9. **Threshold & Fallback:** Return the answer if the score meets the threshold; otherwise, show a fallback response.
10. **Chat & Storage:** Display the response through Stream lit and save conversation history using SQLite.





TOOLS & TECHNOLOGY

1. **Python** – Main programming language used to develop the Chabot.
2. **Pandas** – Used for CSV loading, data cleaning, and dataset handling.
3. **NumPy** – Used for numerical operations and finding the highest similarity score.
4. **Sentence Transformers** – Used for generating semantic embedding.
5. **all-MiniLM-L6-v2** – Retrained model used for sentence-level semantic representation.
6. **Scikit-learn** – Used to calculate cosine similarity.
7. **Streamlit** – Used to create the interactive Chabot web application.
8. **SQLite** – Used for persistent storage of chat history.
9. **CSV** – Used as the educational knowledge-base format.
10. **VS Code / Python Environment** – Used for project development and testing.





FUTURE ENHANCEMENT

- 1. Larger Knowledge Base & Document Retrieval**
– Add PDFs, notes, textbooks, and other educational documents.
- 2. Hybrid Search & RAG** – Combine semantic search with keyword search and Retrieval-Augmented Generation for better answers.
- 3. Multilingual & Voice Support** – Allow students to ask questions in multiple languages using voice input.
- 4. Cloud Deployment & User Management** – Add authentication, cloud database, analytics, and deploy the Chabot for multiple users.





THANK YOU